UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 002 · Solutions of homogeneous DE: complementary functions · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
added for clarity	Standard context that is not in the class material
transcript only	Said in the lecture but not on any slide
video only	Seen only in a frame of the class video
not on current slides	From older material or past papers, not this year's slides
S002·7	Unit 2 deck 002, slide 7 (rendered .pptx)
T002 00:09:10	Unit 2 AV-summary transcript 002 (“Rules for finding the Complementary Function, Problems”), segment start
CW U2 P8	Unit 2 class-work problem 8
ESA Jan-24 s4 2c	Past paper: UE23MA141A Jan 2024 set 4, question 2.c

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4. Figure G3.2 · A repeated complex pair: $\cos 5x$ against $x \cos 5x$
5. Figure G4.1 · The solution of CW P9 meets its four conditions

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = first solution, vermilion (dashed) = second, green = third, purple = fourth; grey dashed lines are references. Every plotted solution was found and checked with sympy.

2 Second-order equations: three cases

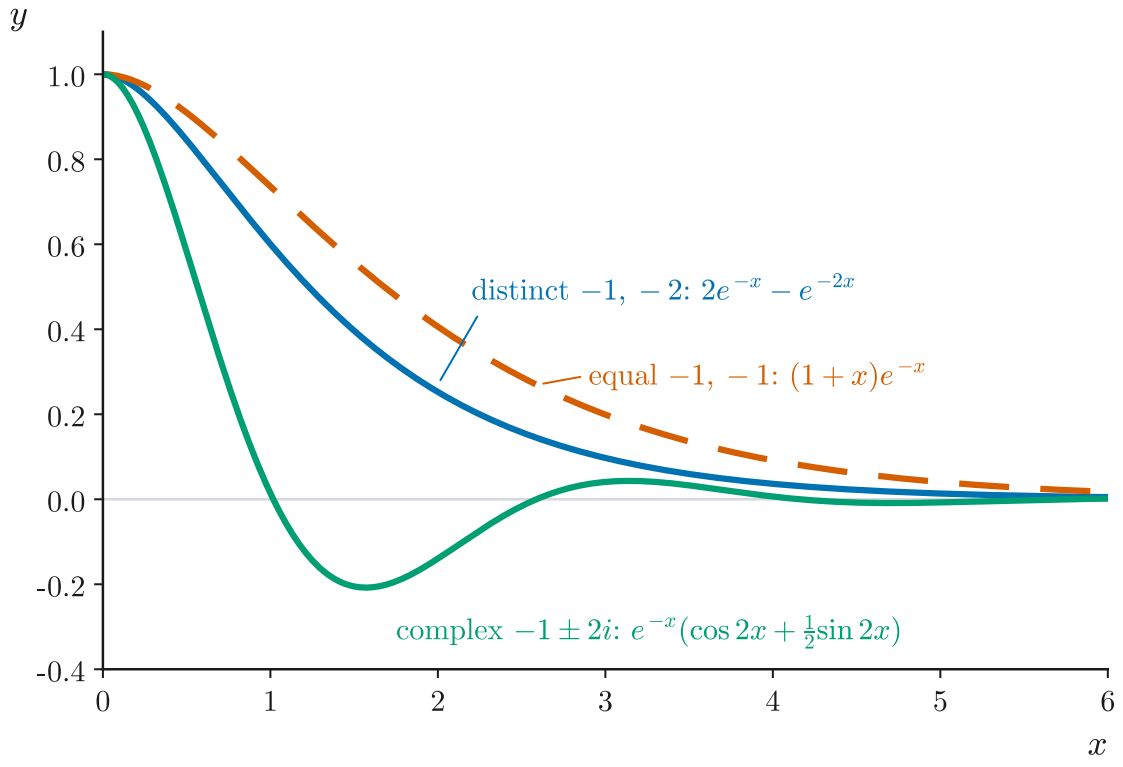


Figure G2.1. The three second-order cases from the same start. ESA ★★★

ADDED FOR CLARITY S002·7 · S002·8 · S002·9

How to read it. Each curve solves a second-order equation with $y(0) = 1$, $y'(0) = 0$. Distinct roots $-1, -2$ ($y'' + 3y' + 2y = 0$, blue) and the double root -1 ($y'' + 2y' + y = 0$, vermilion dashed) decay without crossing zero; the double root is the fastest return that does not oscillate. Complex roots $-1 \pm 2i$ ($y'' + 2y' + 5y = 0$, green) oscillate inside the decaying envelope e^{-x}. Each solution was found and checked with sympy.

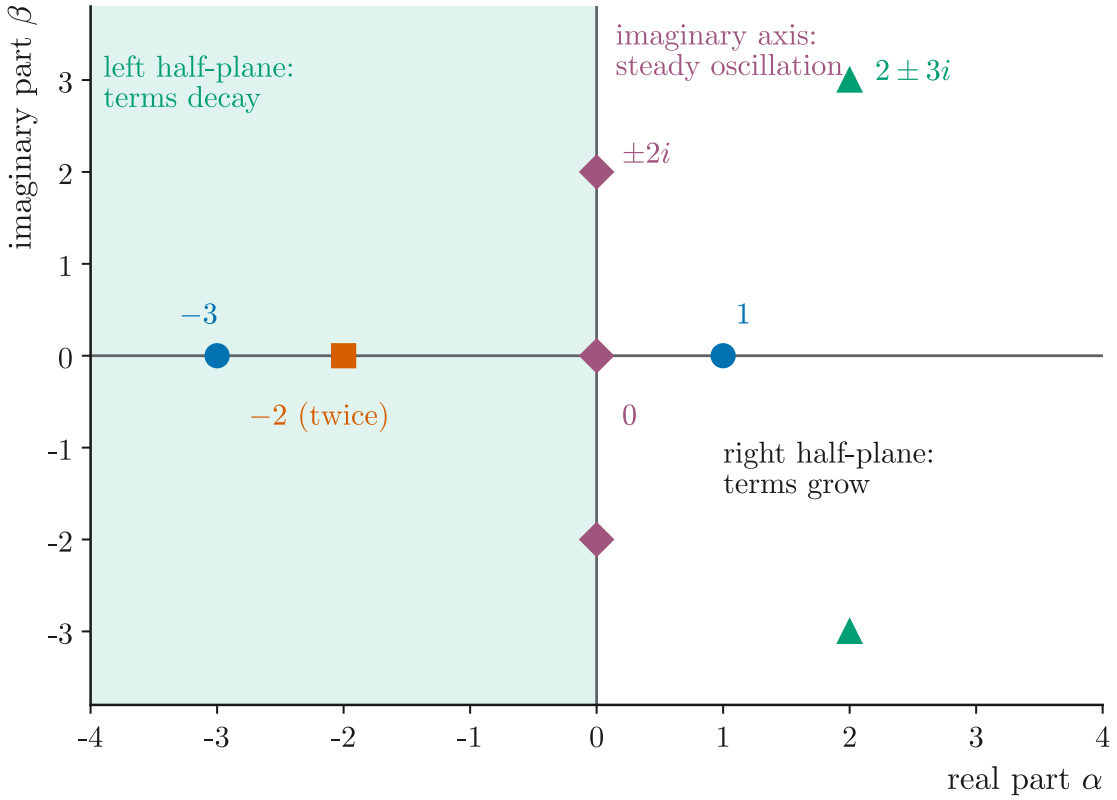


Figure G2.2. Where the roots lie decides the shape of the solution. ESA ★★

ADDED FOR CLARITY S002·7 · S002·8 · S002·9 · S002·14

How to read it. The roots of the four slide examples in the complex plane: $-3, 1$ (slide 002·7, circles), -2 twice (slide 002·8, square), $2 \pm 3i$ (slide 002·9, triangles), $0, \pm 2i$ (slide 002·14, diamonds). The real part α gives growth ($\alpha > 0$) or decay ($\alpha < 0$, green half-plane); the imaginary part β gives oscillation $\cos \beta x$, $\sin \beta x$. Roots on the real axis give no oscillation.

3 Equations of order n: combining the cases

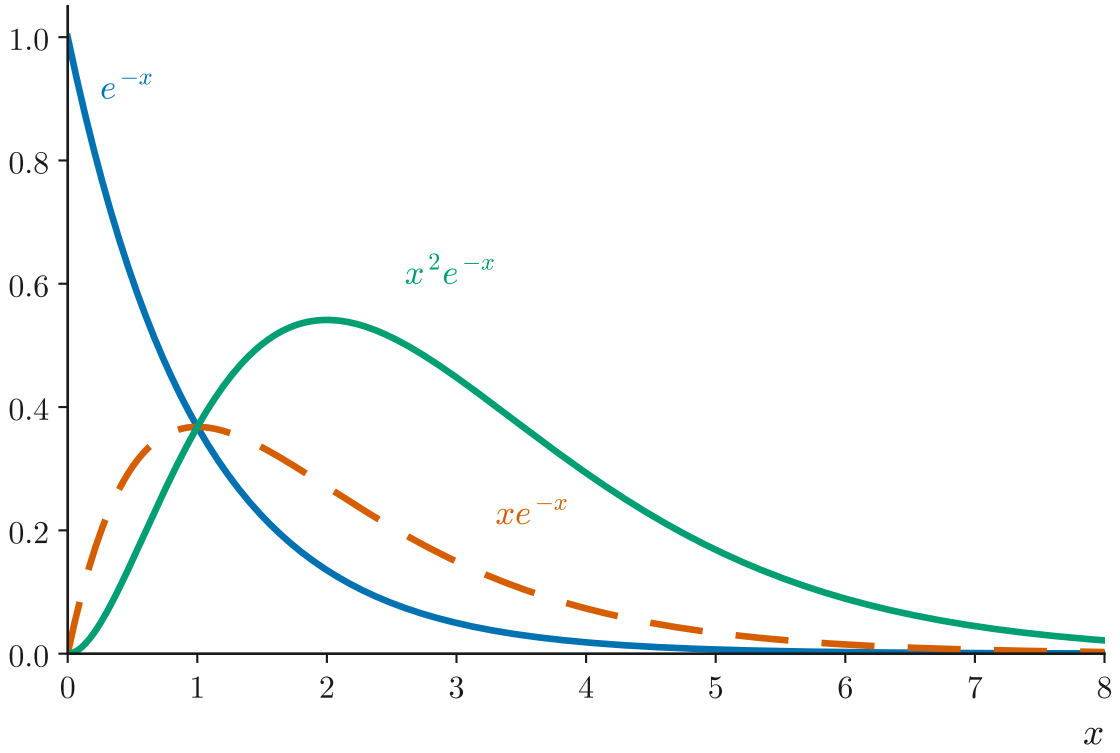


Figure G3.1. A triple root $m = -1$: the three solutions $x^k e^{-x}$. ESA ★★

ADDED FOR CLARITY S002·12 · T002 00:15:15

How to read it. For $(D+1)^3 y = 0$ the CF is $(c_1 + c_2 x + c_3 x^2)e^{-x}$. The three curves e^{-x}, xe^{-x}, x^2e^{-x} have different shapes (peaks at $x = 0, 1, 2$), so they are independent and give three constants. All three still decay because $m < 0$.

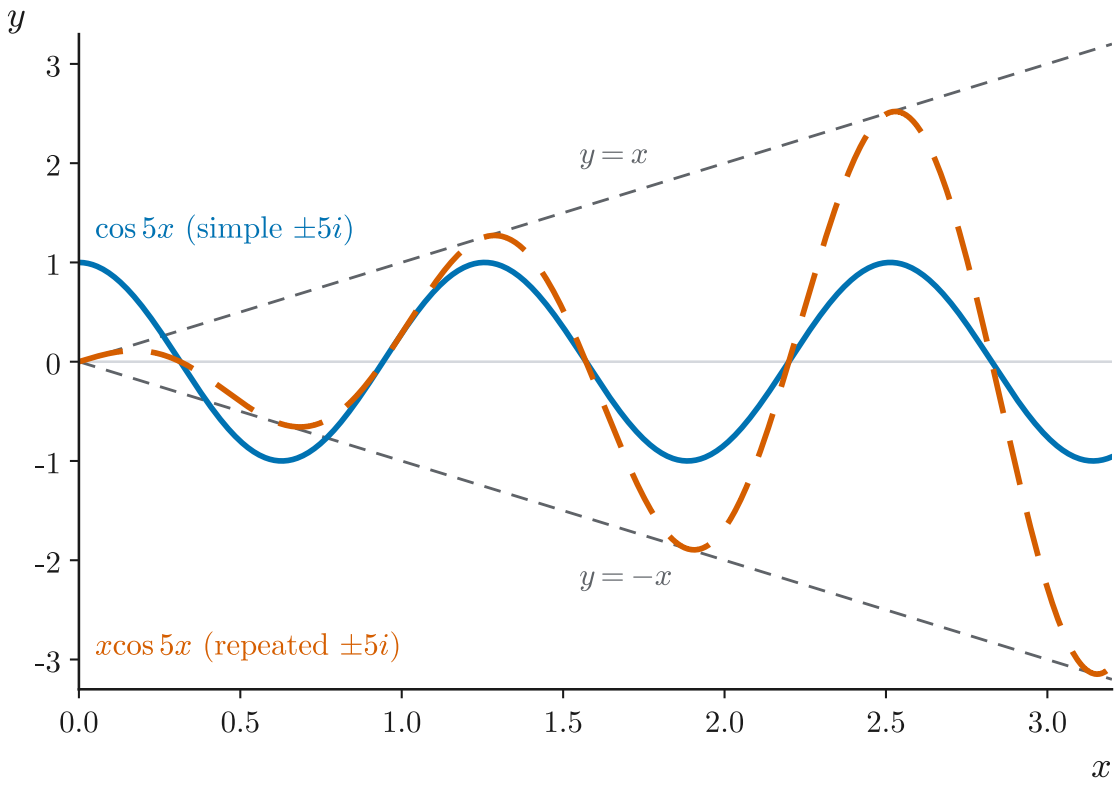


Figure G3.2. A repeated complex pair: $\cos 5x$ against $x\cos 5x$. ESA ★★

ADDED FOR CLARITY CW U2 P7 · S002·13

How to read it. For CW P7 the roots $\pm 5i$ occur twice. A simple pair would give the bounded $\cos 5x$ (blue); the repeat adds $x\cos 5x$ (vermilion dashed), whose amplitude grows linearly between the grey lines $y = \pm x$, although $\alpha = 0$.

4 Solving problems, with initial conditions

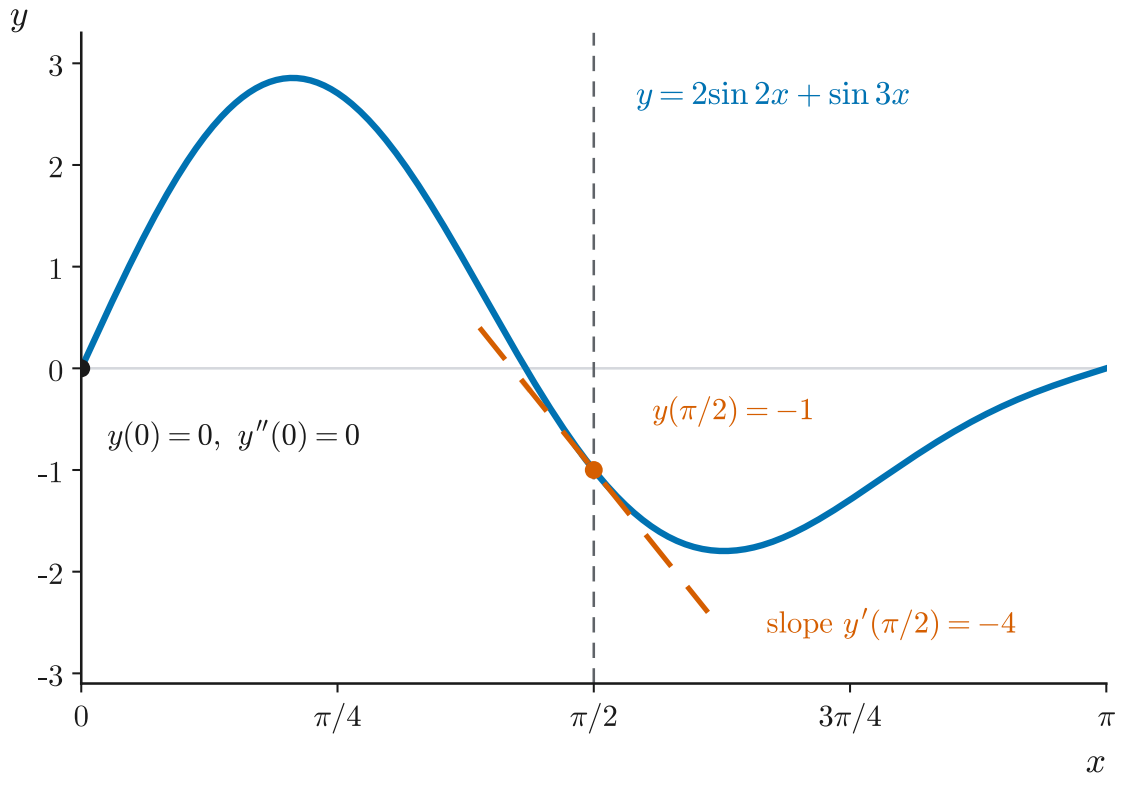


Figure G4.1. The solution of CW P9 meets its four conditions. ESA ★★

ADDED FOR CLARITY CW U2 P9

How to read it. $y = 2\sin 2x + \sin 3x$ (blue) passes through the origin with $y''(0) = 0$ (black dot) and through $(\frac{\pi}{2}, -1)$ (vermilion dot) with slope -4 (vermilion dashed tangent). Four conditions fix the four constants of a fourth-order equation.